Venkata Lakshmi Parimala Pasupuleti

AI Engineer · Clinical AI · Ambient AI · Agentic Workflows · AI Skills Library · Python · TypeScript · React · LangGraph · FHIR · EHR Integration · HIPAA · LLM APIs · Production AI

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PROFESSIONAL SUMMARY

Fullstack AI engineer building agentic clinical AI systems — with 8 years of healthcare software production depth and 2 years hands-on LLM + agentic AI delivery. Built **AI on FHIR**: NLP platform integrating LLMs over FHIR EHR APIs with HIPAA-compliant RBAC, translating clinical queries into structured patient data in real-time. Built **Leasel**: 8-agent LangGraph compliance investigation platform with reusable specialized sub-agents — the same AI Skills Library pattern Commure's Air AI team is building. Built **AgentProbe**: LLM production eval infrastructure with 5-metric engine, CI/CD regression gate, and model routing. Stack: Python (backend) + TypeScript/React (frontend) + Docker/K8s. Design rule enforced in code: LLM reasons, clinician or deterministic logic decides every high-stakes call. Currently building ML pipelines at Blue Horse Digital. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

WORK EXPERIENCE

Blue Horse Digital
Sep 2026 – Present

AI/ML Engineer (Volunteer) · Virginia, USA

- Developing ML models and data analysis pipelines; integrating AI components with back-end and front-end teams; building pipelines that surface business data insights into application APIs and reporting layers. Fast shipping, cross-functional collaboration with product and engineering.

Magizhs Tech Solution
May 2025 – Aug 2025

AI/ML Engineer Intern — Agentic AI · LangGraph · RAG · Production Deployment

- Built agentic compliance system (LangGraph stateful graph + RAG over regulatory corpus) from ideation to production: natural-language input → structured flight plan → validation guardrails → conditional routing to export or rejection. Deployed production client system with CI/CD. Fast shipping, full ownership.

Entrata (PropTech / Real Estate SaaS)
Sep 2018 – Aug 2024

Senior Software Engineer — Full-Stack · Healthcare-Grade Compliance · Python · PHP · React · AWS

- **Shipped 20+ full-stack production modules** across 120+ Agile sprints (Python, PHP, PostgreSQL, JavaScript/React) — at a scale of 100K+ monthly active users at 99.5% uptime. Led microservices redesign with Docker — 50% scalability improvement, 40% deployment reduction. Mentored 5+ engineers, 300+ peer code reviews.
- **Owned HUD compliance pipeline for 7 years** (180,000 domain records for housing eligibility, rent, HAP, tax credits) — healthcare-grade correctness: wrong data = wrong outcomes for real people. Zero compliance violations. RabbitMQ async processing with idempotency guarantees.
- **Led AWS S3 migration with 100% PHI compliance** — Libsodium-encrypted pipelines, zero security incidents. Optimized 200+ SQL queries (30% gain). Built internal debugging tools reducing bug count below 10 for 6+ consecutive months.

George Mason University
Aug 2024 – Aug 2025

Research Associate & Teaching Assistant

- Built ML models ($R^2 = 0.977$), deployed inference API (HuggingFace + PyTorch + Quart async). Anomaly detection (Isolation Forest) across 3 data pipelines. TA for 600+ students: Python, OOP, SQL.

Entrata India, Pune
Jan 2017 – Sep 2018

Associate / Trainee Software Engineer

- Full-stack modules (PHP, PostgreSQL, JavaScript); AWS S3 migration; redesigned document storage architecture for hundreds of thousands of records.

PROJECTS

AI on FHIR: Clinical NLP Platform — EHR Integration, HIPAA RBAC, Multi-Language
CLINICAL AI · EHR · FHIR

Flask · LLMs · FHIR API · BERT · Embeddings · spaCy · React · Docker — github.com/venkatalakshmiparimala/AI_on_FHIR

- ~60% reduction in manual clinical retrieval time — **NLP platform translating natural-language queries (EN/ES) into FHIR EHR API calls** using LLMs, BERT, and embeddings across structured medical records; enforced **HIPAA-compliant RBAC** with data isolation. Frontend in React. Same architecture as a conversational clinical assistant grounded in live patient context.

Leasel: AI Skills Library — Reusable Specialized Agents, HITL, Compliance Workflows
AI SKILLS LIBRARY · AGENTIC AI

Python · LangGraph (8-agent) · FastAPI · ChromaDB · OCR · SQLite — github.com/venkatalakshmiparimala/Lease_complaince_raq

- 8 reusable specialized agents (Triage, Config, Historical, StructuredData, Reasoning, Validation, Human Escalation) assembled into clinical-grade investigation workflows with **confidence-based HITL** — same AI Skills Library pattern. Core rule: **LLM reasons; clinician or deterministic logic makes every high-stakes call**. Eval: **93.6% precision, 87.9% F1, pass@3 88.3%** across 1,240 compliance PDFs.

AgentProbe: LLM Eval Infrastructure — CI/CD Regression Gate, Model Routing, Observability
LLM EVAL · PRODUCTION AI

Python · FastAPI · LangGraph · OpenAI · Anthropic · Groq · Ollama · CI/CD (GitHub Actions) — github.com/venkatalakshmiparimala/AgentProbe

- 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression harness, model routing across OpenAI/Anthropic/Groq/Ollama, offline + online evaluation loops. Establishes "ships with quality" culture — quality gates on every commit before patient-facing features deploy.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

AI systems, ML modeling, clinical data engineering, NLP, statistical modeling

B.Tech Computer Science & Engineering — JNTU Anantapur — 2013 – 2017

TECHNICAL SKILLS

Clinical & AI: Clinical AI, Ambient AI, EHR Integration, FHIR, HIPAA Compliance, Clinical NLP, Clinical Workflows, Patient Workflows, HITL, LangChain, LangGraph, Agentic AI, Multi-Agent Orchestration, AI Skills Library, RAG Pipelines, Function Calling, Structured Outputs, LLM Evaluation, LLM-as-Judge, Hallucination Detection, Confidence Scoring, Model Routing, Agent Observability, Prompt Engineering, OpenAI API, Anthropic API, Groq, Ollama, ChromaDB, Qdrant, Production AI

Full-Stack: Python (backend, expert), TypeScript, React.js, JavaScript, PHP, FastAPI, Spring Boot, REST APIs, HTML/CSS; Docker, Kubernetes, CI/CD (GitHub Actions), Microservices, Distributed Systems, RabbitMQ, Redis, AWS (S3, EC2, RDS), Azure (ADF, ADLS, Databricks), GCP

Data & ML: PostgreSQL, MySQL, Oracle, Snowflake, NoSQL; PyTorch, TensorFlow, Scikit-learn, HuggingFace, XGBoost, Isolation Forest, Pandas, NumPy, spaCy, BERT; ETL, Data Pipelines, Apache Spark, PySpark; SQL, PL/SQL, R, Bash, Scala, Java